

Meterpreter vs Reverse Shell — Decision Guide

Understand what decides the session type, what LHOST/LPORT mean, whether port 443 changes anything, transport choices, session management, and upgrade decisions.

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Usage note: For authorized eJPT-style labs. Use the MSFvenom guide for payload generation instead of duplicating large command sets here.

PAYLOAD

Session type

LHOST

Your reachable address

LPORT

Listener port

SESSION

Manage

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1. Key correction: LPORT 443 does not mean Meterpreter

The payload type decides the session; the port is only the connection point.

Payload example	Result	Transport
windows/x64/meterpreter/reverse_tcp	Meterpreter	Reverse TCP
windows/x64/meterpreter/reverse_https	Meterpreter	Reverse HTTPS
windows/x64/shell/reverse_tcp	Normal shell	Reverse TCP

Note: Setting LPORT=443 does not turn shell/reverse_tcp into Meterpreter. To get Meterpreter, the payload itself must be a Meterpreter payload.

2. What do LHOST and LPORT mean?

Option	Meaning	Note
LHOST	Your address the target can reach	In lab VPNs, often the correct tun0/VPN address.
LPORT	The port on which the handler/listener receives the callback	4444, 443, or another lab-appropriate port.
RHOST/RHOSTS	Target(s)	Used by many modules, not every payload.

KALI TERMINAL — Show addresses

```
ip addr
```

If LHOST is not reachable from the target, the session will not return even if everything else is correct.

3. Is port 443 “better”?

Point	Explanation	Result
Possible benefit	Networks often permit outbound HTTPS/443	May help in some authorized lab scenarios.
Not automatically HTTPS	reverse_tcp on 443 is still TCP payload traffic	The port number does not change the protocol.
Actual HTTPS	Use a reverse_https payload	Payload type determines this.
Low port on Kali/Linux	Binding below 1024 may require privileges/capability	May need suitable permissions or another port.

Note: Do not use 443 or payload choices to bypass controls on networks you are not authorized to test.

4. reverse_tcp vs reverse_http vs reverse_https

Type	Idea	When to consider it
reverse_tcp	Simple direct TCP	Internal labs and direct connectivity.
reverse_http	HTTP transport	When the lab scenario is designed for HTTP.
reverse_https	HTTPS transport	When the scenario requires HTTPS or proxy-friendly transport.

Do not switch transports simply because one sounds “stronger.” Start with what the lab requires and the simplest working connection.

5. Meterpreter commands that save time

METERPRETER

```
getuid  
sysinfo  
pwd  
ls  
cat [FILE]  
ipconfig  
route  
ps
```

METERPRETER — Native system shell

```
shell
```

METERPRETER — Return to Metasploit without killing session

```
background
```

MSFCONSOLE — Sessions

```
sessions  
sessions -i 1
```

METERPRETER — Lab file transfer

```
upload [LOCAL_FILE] [REMOTE_PATH]  
download [REMOTE_FILE] [LOCAL_PATH]
```

6. I have a normal shell - can I upgrade it?

Metasploit can attempt an upgrade in some circumstances.

MSFCONSOLE

```
sessions  
sessions -u 1
```

Replace 1 with the shell-session ID. Success depends on OS, architecture, connectivity, and available payload behavior.

A failed upgrade does not make the shell useless. Continue with it or improve a Linux shell with a PTY when appropriate.

7. Decision Guide

Situation	Choice	Principle
Need only a few commands	Normal shell is enough	Improve with PTY on Linux only if needed.
Need sysinfo/upload/download/session management	Meterpreter can be useful	Use an appropriate Meterpreter payload for the lab.
Lab connection is designed for HTTP/HTTPS	Consider reverse_http/reverse_https	Match handler and scenario.
A shell already works	Do not change it only because Meterpreter looks nicer	Continue when it meets the objective.

8. Common reasons a session does not return

Problem	Sign	First check
Wrong LHOST	Address is not reachable	Check tun0/eth0 and target route.
LPORT in use	Listener cannot bind	Use ss -lntp on Kali.
Port 443 without permission	bind permission denied	Use suitable lab permissions or choose another port.
Wrong Payload / Architecture	OS or x86/x64 mismatch	Review OS/architecture.
Firewall / Egress	Callback cannot leave	Use only connection options authorized by the scenario.

Prepared as a quick-reference companion to the canonical Arabic guide. Use only in labs or systems you are explicitly authorized to test.

References:

[Rapid7 Metasploit Documentation](#)

[Metasploit Framework payload/session documentation](#)